

Kiet Vu

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Education

Purdue University, B.Sc. in Computer Engineering
West Lafayette, IN

Expected Dec. 2028

GPA: 3.93/4.0

Expected Coursework: ASIC Design Laboratory, Microprocessor Systems and Interfacing, Computer Design and Prototyping, Semiconductor Devices, Advanced C Programming

Projects

GPU Research | *Accel-Sim/GPGPU-Sim, CUDA, Nsight Tools, PTX/SASS*

June 2026 – Present

- Restored PTX-mode simulation on all CUDA 11+ toolchains by fixing a parser bug, and added an L2 write-hit counter comparable to NVIDIA's profiler; **both merged upstream**
- Root-caused a SIGSEGV in the FP64 pipeline model (out-of-bounds index when a functional unit's initiation interval exceeds its latency) and fixed a correlation bug that mis-attributed local-memory traffic, cutting L2 write error on one benchmark from **-94.0%** to **+2.6%**
- Investigated an unmodeled **4x over-count** in L1-to-L2 write traffic; built a microbenchmark to confirm the merging mechanism; prototyped a merge buffer, reducing mean L2 write error from **19.48%** to **2.40%** and L2 write-hit error from **19.10%** to **2.33%** across 32 benchmarks, with 19 benchmarks bit-identical

32-bit RISC-V Pipelined Processor | *SystemVerilog, Computer Architecture*

Feb. 2026 – Mar. 2026

- Architected a 5-stage pipelined RV32I processor using Harvard memory architecture with a Hazard Control Unit; supported the complete base integer instruction set, achieving **100% architectural compliance** via the RISCOF framework by co-simulating against Spike reference model
- Synthesized the core targeting the Sky130 node via LibreLane to generate a verified GDSII layout; optimized PPA trade-offs using high-speed standard cells to achieve a peak frequency of **95.2 MHz (0.551 mm²)**, alongside a high-density variant reaching **58.8 MHz (0.427 mm²)**, both with clean STA and LVS signoffs

Jane Street ASIC Reverse Engineering | *Verilog, Formal Verification, RTL-to-GDS tools*

Aug. 2026

- Recognized by Jane Street as one of their favorite submissions to the 2026 ASIC reverse-engineering puzzle
- Reverse-engineered the architecture of an 11×11 constraint validator with 22 two-bit column/region accumulators and a 12-stage adjacency delay line via cone-of-influence tracing, placement clustering, and a 121-case perturbation sweep that exposed mod-11 raster periodicity
- Rebuilt the design forward from RTL-to-GDS with **0 DRC/LVS/antenna violations** and setup closed on all 9 corners, then proved output equivalence to the extracted original with SAT miters across all 2¹²¹ input boards

Experience

Undergraduate Researcher, Purdue SoCET GPU Team

Aug. 2026 – Present

- Architected the Ethernet RX datapath for a fixed-function GPU on FPGA, produced RTL block diagrams and state machines for frame detection, CRC-32 validation, address filtering, and CDC buffering
- Designed a custom host-to-GPU command protocol over raw Ethernet with register-write and vertex-stream

Undergraduate Researcher, Purdue EPICS HDR

Aug. 2025 – May 2026

- Refactored multiple legacy data source modules to standardized abstract base class architecture, implementing consistent data pipeline, improving maintainability and code reusability
- Eliminating NULL overwrites and integrating 1,800+ historical records into SQLite; centralized configuration consolidating 59 SQL field mappings, 1,000+ station IDs, and 7 database schemas across 7 data sources for 53+ monitoring locations

Extracurricular Activities

Technical Member, Purdue RoboMaster

Aug. 2025 – May 2026

- Engineered custom PCBs for a rechargeable armor plate and fluorescent projectile charger, integrating TP4056 chargers, TPS63002 buck-boost converters, ESP32 monitoring, and MOSFET/Zener-based battery protection
- Engineered an audio visualizer circuit, amplifying a microphone signal by 45x with a cascaded LM358 op-amp and driving a 10-segment LED display with an LM3915 IC

Technical Skills

Languages: Verilog/SystemVerilog, C/C++, Tcl, CUDA, PTX/SASS, MATLAB, Python

Computer Architecture: GPU architecture, RISC-V, I/O Interface, Ethernet, AXI, AMBA, FPGA

Software: LibreLane Toolchains, Yosys, Surfer, Linux, GitHub, EasyEDA, Altium, VS Code, Nsight Tools